AD DS FOUNDATIONS

One directory. Central management.

1 Topology

The physical or logical layout of a network.



2 Workgroup

Each computer manages its own local accounts.



3 Domain

Centralized identity and administration for joined computers.



4 AD DS = Active Directory Domain Services.

Stores and organizes users, computers and groups.



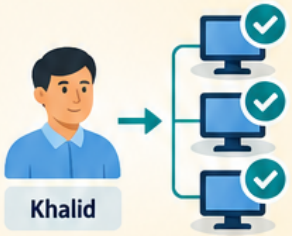
5 Domain Controller (DC)

A server running AD DS that holds directory data and authenticates domain identities.



6 Example

Create Khalid's domain account once. Use it on domain computers where logon is allowed.



7 Key Distinctions

Authentication: Who are you?

Authorization: What may you do?

A domain account does not automatically grant access to every server or folder.

DATABASE & PARTITIONS

What AD stores — and where

1 Schema



Defines object types and attributes.

Example:
user objects can have a telephone number.

2 Configuration



Forest-wide directory structure, sites, subnets and replication topology.

3 Domain



Directory objects for a domain, such as users, groups and computers.

4 Application



Service-specific directory data.

Example:
AD-integrated DNS records.

Replicates to selected directory replicas.

5 NTDS.DIT



AD directory database.

Default:
`C:\Windows\NTDS\ntds.dit`

6 SYSVOL



Group Policy files and logon scripts.

Default:
`C:\Windows\SYSVOL`

7 Transaction logs



Record database changes and support crash recovery.

Default locations can be changed during setup.
Transaction logs are not user logon history.

KERBEROS AUTHENTICATION

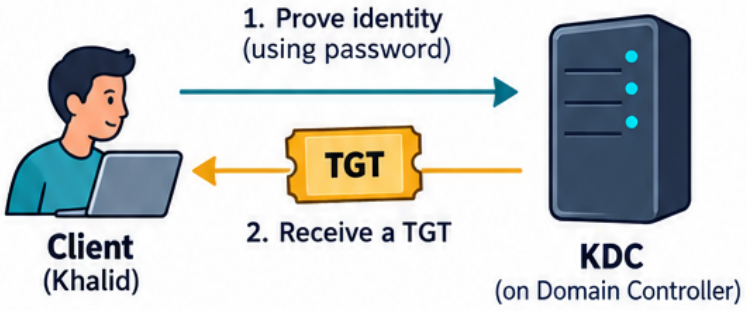
Identity through tickets

KDC = Key Distribution Center.
A service on a Domain Controller.

1 Sign in

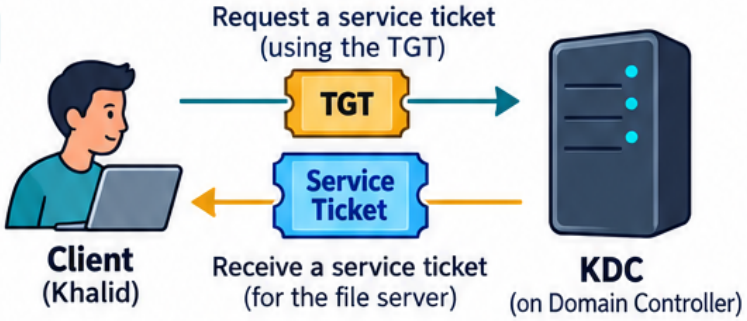
The client proves the user's identity to the KDC and receives a TGT.

TGT = Ticket Granting Ticket.



2 Request a service

The client uses the TGT to request a service ticket for the file server.



3 Access the resource

The client presents the service ticket. The file server validates it and checks permissions.



Example:
Read permission: Khalid can read the file.
Without Write permission, he cannot edit it.



✓ **Read** – **Allowed**
✗ **Write** – **Not allowed**

Tickets have a limited lifetime.
TGTs and service tickets expire after a period of time.

The user's password is not sent to each resource server.
After sign in, the client uses tickets instead of the password.

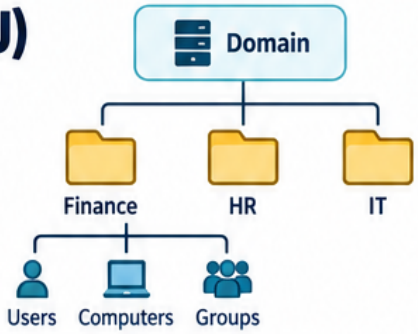
Kerberos verifies identity. Permissions control access.

ORGANIZE, CONTROL & GROW

OU • Groups • Policies • Delegation

1 Organizational Unit (OU)

A directory container for users, computers, groups and other OUs. Organizes objects, supports Group Policy links and delegated administration.



- Organizes objects
- Supports Group Policy links
- Enables delegated administration

2 Security group

Collects members so resource permissions can be assigned to the group.
A user can belong to several groups.



Example

- Add users to a group
- Assign permissions to the group
- Users can be members of several groups

3 Policy vs permission

Policy

Configures behavior, such as automatic screen locking.

Permission

Allows an action, such as reading a folder.

Important

GPO settings apply to users or computers according to setting type, scope and filtering.

Groups do not receive GPO settings directly.

4 Scalability

Support more users, computers and directory workload as the organization grows.

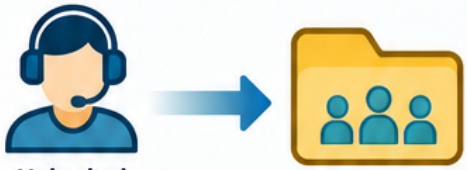


Example

- More users
- More computers
- Higher directory workload
- Grows with organizational needs

5 Delegation

Give limited administrative rights for a specific task. Example: help desk can reset passwords for users in the Finance OU without becoming Domain Admins.



Department transfer: remove old group membership and add the appropriate new membership. Access changes may require a refreshed sign-in token.